

- Q.29 Discuss various air safety requirements and standards.
- Q.30 Draw equilibrium of forces on wing and tail.
- Q.31 Describe the function of an Aileron and rudder?
- Q.32 Enumerate the aerodynamic forces acting on aircraft.
- Q.33 Explain Gas turbine jet engine working in brief.
- Q.34 Explain about the Hydraulic systems used in aircraft.
- Q.35 Explain the term air worthiness.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Describe the working of rate of climb indicator. What are the other pilot based instrument used?
- Q.37 Describe the basic criterion for the material selection for various aircraft parts, describe the properties of each material as well?
- Q.38 Write a short note on any two-
- Functioning of aircraft electric system and autopilot
 - Types of drag acting on aircraft
 - Aircraft fuel system

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3rd Sem / Aircraft Maintenance Engg. Sub : Introduction to Aeronautics

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which component of a hydraulic system is responsible for converting hydraulic energy into mechanical motion?
- Accumulator
 - Actuator
 - Reservoir
 - Filter
- Q.2 Which component of a pneumatic system is responsible for removing moisture and contaminants from the air before it enters the system?
- Valves
 - Filters
 - Capsules
 - Bellow
- Q.3 Major development in aircraft field took place in?
- 1920s
 - 1940s
 - 1990s
 - 2000s
- Q.4 Which engine has the highest efficiency?
- Turbo Jet Engine
 - Turbo Fan Engine
 - Rocket Engine
 - Turbo shaft Engine

- Q.5 If an airfoil is operating at Mach 1.1 then it is working in
 a) Transonic range b) Supersonic Range
 c) Subsonic range d) Hypersonic range
- Q.6 What is bleed Air?
 a) Air around the tail
 b) Leaking air of the engine
 c) Air taken from compressor
 d) None of the above
- Q.7 The cross sectional shape of wing is called
 a) Airfoil b) Circle
 c) Camber d) Chord
- Q.8 The Pitot tube is used to measure
 a) Velocity at stagnation point
 b) Stagnation pressure
 c) Static Pressure
 d) Dynamic Pressure
- Q.9 Which body issues Civil Airworthiness requirement in India
 a) FAA b) EASA
 c) DGCA d) ICAO
- Q.10 What is flight control system
 a) Autonomous flight system
 b) Air traffic control system
 c) Engine throttle control
 d) System to move control surfaces

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 What is primary control?
 Q.12 How de-icing is done?
 Q.13 How does a Hovercraft work?
 Q.14 Name one pneumatic system task in an aircraft.
 Q.15 What is critical Mach number?
 Q.16 What is wing loading?
 Q.17 Expand D.G.C.A.
 Q.18 What is the first estimation taken for aircraft design?
 Q.19 Name various engines used in aircraft?
 Q.20 Name fuselage components?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Describe about three primary controls of an aircraft?
 Q.22 Explain about basic components of Piston Engine.
 Q.23 Write a short note on V/STOL machines.
 Q.24 Draw pressure distribution curve over an airfoil.
 Q.25 Explain basic steps in airplane design?
 Q.26 Draw shear force diagram of a rectangular wing?
 Q.27 Describe in brief about Pneumatic system used in aircraft?
 Q.28 Discuss the statement "Every gram counts".